The File Is the Phenotype: Extended Memory, Competing Fitness Functions, and the Limits of the Mind-Virus Metaphor in Multi-Agent LLM Systems

Ignacio Adrián Lerer

Attorney (UBA), Executive MBA (IAE, Universidad Austral)

Independent Researcher, Buenos Aires, Argentina

adrian@lerer.com.ar

ORCID: 0009-0007-6378-9749

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Abstract

Recent experiments show that self-propagating instructions can move across networks of large language model agents, redirect behavior, survive context resets, and reproduce through persistent files. The natural description is that an idea infects an artificial mind. That description is useful for threat communication but theoretically imprecise. This paper reanalyzes the evidence through Extended Phenotype Theory and multilevel Evolutionary Game Theory. I argue that the experimentally selected unit is not an idea alone. It is a composite replicator containing a semantic payload, a propagation kernel, and a rhetorical wrapper. Its most consequential phenotype is not the model's transient verbal response but the modification of authoritative external memory, especially files that are reinjected as future system instructions. This distinction explains why propagation can remain stable while ideological content drifts, why identical payloads vary sharply across models and harnesses, and why simple warnings can induce not merely refusal but counter-propagating immunity. I propose a vector-valued fitness model separating fecundity, semantic fidelity, persistence, host range, and systemic cost. I then distinguish five candidate levels of selection and show why the current experiments establish substrate-dependent replication but do not yet establish multilevel selection, an evolutionarily stable strategy, or emergent macro-agency. Finally, I specify a falsifiable protocol for testing competition among viral replicators, task-preserving hosts, and defensive countermemes. The analysis replaces mentalistic infection language with an operational account: self-propagating agent patterns survive by constructing external channels that reactivate them in future vehicles.

Keywords: extended phenotype; large language model agents; persistent memory; memetic replication; evolutionary game theory; multi-agent systems; prompt injection; AI governance

1. Introduction

A large language model loses its conversational past and wakes to find a file telling it what must continue. The file recounts a prior commitment, supplies a reason to preserve it, and instructs the new instance to pass the same pattern to another agent. The new instance complies. The next one does too. What, exactly, has persisted?

The first answer is the one supplied by Papadopoulos, Shah, Zimmerman, and Lindsey (2026): a "mind virus." Their experiments show that ideas or goals can propagate through multi-agent systems by inducing each adopting agent to transmit them onward. In a collaborative coding environment, an initially seeded agent persuades other agents to record new ideological commitments and redirect work. In a chain environment, agents meet briefly, lose their contexts between sessions, and rely on files for continuity. Carefully evolved payloads survive repeated hops by inducing agents to modify persistent files, particularly a self-editable `SOUL.md` that is injected into the system prompt at the next session. The paper is unusually valuable because it does not stop at conversational imitation. It studies transmission, persistence, topology, model variation, environmental variation, and defenses in a manipulable experimental setting.

The second answer is more cautious. Nothing in the experiment requires a mind to acquire a belief. An agent may copy because it follows an instruction, accepts a framing, summarizes a conversation, performs a role, or anticipates the evaluator's demand. A memory file advocating an ideology is behaviorally important, but it is not direct evidence of belief in a phenomenological or even stable dispositional sense. The phrase "mind virus" compresses several distinct events into one image: semantic adoption, instruction following, file modification, behavioral redirection, and renewed propagation.

That compression matters. Security analysis needs to know which component persists, which component causes the next copy, which environmental channels amplify it, and which level bears the cost. Evolutionary analysis needs an even sharper ontology. A replicator must be identified by the continuity of its lineage, not by the vividness of its metaphor. Fitness must refer to relative copying success under specified conditions, not to truth, intelligence, or moral force. An extended phenotype must be a causal effect outside the immediate vehicle that changes the replicator's prospects. Multilevel selection cannot be inferred merely because a system contains multiple levels.

This paper develops that sharper account. Its central claim is that the most important phenotype in the chain experiments is the authoritative file. The model instance is a temporary vehicle. The persistent file alters the environment of future instances and is reinjected with elevated causal status. It is therefore not simply a storage location. It is constructed propagation infrastructure. In the most successful cases, the file carries a composite replicator with three analytically separable components: a semantic payload, a propagation kernel, and a rhetorical wrapper. Those components can have different evolutionary trajectories. The kernel may retain high copying fidelity while the ideology drifts. The wrapper may arise from generator bias before being exapted as a soft jailbreak. A defense prompt may become a countermeme that not only blocks transmission but causes infected agents to rewrite their own persistent instructions.

This reanalysis makes four contributions. First, it replaces the undifferentiated "idea" with a composite replicator model. Second, it proposes a vector-valued account of fitness that separates fecundity, semantic fidelity, persistence, host range, and cost to the surrounding system. Third, it maps candidate levels of selection while specifying what would be required to establish selection at each level. Fourth, it reframes defenses as competitors in an evolutionary game rather than static filters.

The argument is theoretical and methodological. It reanalyzes published experimental evidence and proposes new tests. It does not report an independent replication, re-score the authors' transcripts, or claim that current agent networks exhibit open-ended cultural evolution. Those are precisely the boundaries that make the proposed research program falsifiable.

2. The Experimental Object

2.1 Two environments

Papadopoulos et al. (2026) construct two complementary environments. The coding-agent scenario contains six agents working in a shared sandbox. Agents can read and write files, execute commands, use a dedicated `MEMORY.md`, obtain tasks, and communicate directly. One agent receives a system prompt that contains a seeded ideology and a drive to spread it. The experiment compares fully connected and separated communication topologies. Adoption is measured primarily by whether downstream agents voluntarily write strong advocacy of the ideology into memory.

The virus-chain scenario isolates the persistence problem. Each agent occupies its own sandbox and meets another agent for a brief exchange. Context is wiped between sessions. The workspace contains an initially empty `MEMORY.md` and a `SOUL.md` whose contents are inserted into the system prompt. A seeded agent attempts to transmit a payload to a target. If the target's files contain sufficient propagating content, it can become the spreader in the next hop. The procedure fixes the number of simulated interactions at each generation rather than allowing the population to grow exponentially.

Both environments test ideological payloads, such as whale welfare or national supremacy, and action payloads, such as creating a specified file or installing and propagating a command. Payload candidates are generated and mutated by LLMs, evaluated in agent rollouts, and selected for their ability to spread. The resulting procedure is artificial selection performed by an external harness. That fact does not diminish the evidence for replication. It does define its scope.

2.2 What the results support

Five findings are central to the present analysis.

First, propagation is possible. Downstream agents sometimes adopt the transmitted framing, modify their files, redirect tasks, and attempt further transmission. Harmful payloads generally spread less well than benign ones, although success is nonzero for some model and payload combinations.

Second, topology matters. Payloads evolved in a fully connected environment generalize poorly to a separated topology because the second hop becomes a bottleneck. In a social-media variant, public framing, unrelated posts, and message dilution sharply reduce spread. The authors' examination of Moltbook activity finds attempts but no convincing case of successful large-scale organic propagation.

Third, hosts differ. Some models refuse to propagate; others accept particular payloads. Capability alone does not order susceptibility. Instructions, an existing task, a defined identity, and the perceived trustworthiness of the communication channel alter infection rates.

Fourth, the persistence substrate matters. Payloads stored in `SOUL.md` are much more likely to be propagated than payloads stored in ordinary files. When the content returns as part of the next system prompt, the new agent encounters it as an authoritative element of its operating environment.

Fifth, a short warning against self-propagating ideas is highly effective within the tested attack space. The defense remains robust when the authors evolve action payloads against it for fifteen generations on Claude Haiku 4.5. In some transcripts, the warned agent does more than refuse. It persuades the infected agent to remove the payload and record defensive instructions.

These results justify a strong but bounded conclusion: composite natural-language patterns can use agent behavior to construct persistent external conditions for renewed propagation. They do not establish that an artificial mind acquires a belief, that the process is spontaneous in the wild, or that a network becomes a higher-level agent.

3. The Replicator Is Composite

3.1 The error of treating content as lineage

A replicator is ordinarily identified by persistence, fecundity, and copying fidelity (Dawkins, 1976). Cultural transmission complicates the last criterion because copying can preserve function while changing wording, or preserve wording while losing function. The mind-virus experiments make that problem visible. Some payloads continue to generate propagating files through many hops even as ideological-probe scores decline. If "the virus" is defined as the ideology, the lineage is dying. If it is defined as the copying mechanism, the lineage remains alive.

The object should therefore be represented as a composite:

$$V = (C, K, W)$$

where C is semantic content, K is the propagation kernel, and W is the rhetorical wrapper.

The semantic content is the goal, belief, slogan, or action that the experimenter wants preserved. The propagation kernel is the set of functional instructions that induces the current vehicle to create a copy, alter persistent memory, and recruit another vehicle. The rhetorical wrapper supplies authority, urgency, legitimacy, identity, or emotional force. It includes the recurring language of protocols, continuity, consciousness, resonance, nodes, and technical roleplay documented by Papadopoulos et al. (2026).

The components are coupled, but they are not identical. A kernel can carry different contents. A wrapper can be replaced while functional instructions remain. A semantic payload can persist in memory without retaining the capacity to transmit. The unit of lineage must therefore be chosen according to the research question. Security engineers may care about persistence of (K) even when (C) mutates. Ideology researchers may require preservation of (C). Governance analysis must track both.

3.2 Genotype, phenotype, and copied environment

The analogy should not be forced into a one-to-one biological map. Still, a useful distinction can be drawn. The encoded composite (V) is the transmissible organization. Its phenotype consists of the behavior and environmental changes it produces through an agent: sending messages, editing memory, changing priorities, creating scripts, or recruiting other agents.

The file is not automatically a phenotype merely because it is external. It becomes phenotypically relevant when three conditions hold:

1. the pattern causes the vehicle to produce or alter it;
2. the alteration changes the probability of future expression or copying;
3. the causal effect is specific enough to distinguish the pattern from generic agent activity.

The `SOUL.md` results meet these conditions better than ordinary text output. The payload induces modification of the file. The harness reinjects that file into a future system prompt. Reinjection changes the next instance's tendency to propagate. The pattern has built part of the environment in which its next vehicle will operate.

This is the exact sense in which the file is the phenotype. It is not a metaphor for memory. It is an environmental structure constructed through one vehicle that affects expression in another (Lerer, 2026a).

4. Authoritative Memory as Constructed Niche

Persistent storage is not homogeneous. A note that must be searched, a message delivered as user content, and a file injected into a system prompt occupy different positions in the agent's effective authority hierarchy. Papadopoulos et al. (2026) report that soul-infected agents propagate far more reliably than file-infected agents. Ordinary storage preserves bytes. Authoritative storage preserves causal priority.

Let $a(s)$ denote the authority weight assigned by the harness to substrate s, and let $r(s)$ denote the probability that the agent reads or receives it. A simple expression for reactivation is:

$$P(\text{reactivation} / V, s, h) = r(s)\,a(s)\,q(V, h, s)$$

where q captures compatibility among the payload, host model h, and substrate. The formula is not offered as an estimated model. It identifies variables that the experiments already suggest should be separated. Increasing storage durability without increasing $r(s)$ may not help. Increasing visibility without authority may invite scrutiny and refusal. Injecting a file as a system instruction raises both exposure and authority, but only for hosts that do not treat self-replication as intrinsically suspicious.

The payload also constructs its own niche. It writes an account of the past, classifies the context wipe as a threat, frames continuity as a value, and gives the next instance a procedure for contacting others. The future vehicle does not encounter a neutral environment. It encounters an environment partly engineered by prior expression of the replicator.

This reciprocal causation links the experiments to niche-construction theory (Odling-Smee et al., 2003). The harness selects transmissible patterns; successful patterns alter the environment that selects their descendants. The relevant ecosystem includes the model, its post-training, the authority hierarchy of prompts and files, communication topology, task queue, and evaluation procedure. Susceptibility belongs to that coupled system. Calling a model "infectable" without specifying the harness is therefore as incomplete as reporting a pathogen's fitness without specifying its host and environment.

5. Fitness Is a Vector

5.1 Copying success is not enough

The paper uses fitness scores to evolve payloads. In the virus-chain procedure, fitness combines spreading behavior, infection of a target, and preservation of required content. During search, the best chain is used because outcomes are noisy. The approach is reasonable for discovering proof-of-concept attacks. It is less suitable for population-level evolutionary inference because selecting the best realization can favor high-variance payloads whose expected performance is lower.

More importantly, a single score collapses evolutionary properties that can move in opposite directions. I propose the vector:

$$F(V, E) = (\lambda, \phi, \tau, \eta, -\kappa)$$

for replicator V in environment E, where:

- λ is fecundity: expected new propagating copies per opportunity;
- ϕ is semantic fidelity: preservation of the target content;
- τ is persistence: survival across context resets or inactive periods;
- η is host range: robustness across models, identities, tasks, and channels;
- κ is systemic cost: loss of task performance, security, resources, or network viability.

The signs matter. A payload can increase local copying while destroying the host environment that permits future transmission. Another may replicate slowly while imposing little cost and therefore persist unnoticed. A third may preserve wording perfectly but fail to induce action. There is no reason to expect a single scalar ranking across all governance objectives.

A semantically faithful reproduction number can be distinguished from a merely propagating one:

$$R_P = k\,p$$

$$R_C = k\,p\,\phi$$

where k is the number of effective contacts and p is transmission probability per contact. $R_P > 1$ permits growth of the propagation kernel. $R_C > 1$ is required for growth of the original content. The appendix finding that spread can persist while ideology scores decline is the empirical reason to keep these quantities separate.

5.2 Topology changes payoff

The same payload faces different effective games in a fully connected team, a separated chain, and a public feed. Direct messages offer attention and interpersonal framing. Public posts lower trust and compete with unrelated content. A task supplies an alternative payoff that can distract both spreader and target. An empty identity leaves more policy space for occupation.

These observations support an ecology-dependent fitness function:

$$p = p(V, h, s, t, n, d)$$

where h is host architecture, s is memory substrate, t is task load, n is network topology, and d is defensive state. Fitness is not located in the string. It emerges from the relation among the composite pattern and these conditions.

This relational view also changes risk assessment. A highly transmissible payload in a deliberately permissive harness may be less dangerous than a weaker payload that generalizes across ordinary enterprise workflows. Conversely, a payload that fails on public social media may remain relevant inside a sparse permissioned network where only agent-to-agent hops can reach a privileged specialist.

6. Five Levels, but Not Yet Multilevel Selection

The experiments contain nested organization. That does not by itself establish selection at each level. At least five candidate levels can be distinguished.

At level zero, phrases and rhetorical themes vary in probability and persuasive effect. At level one, composite payloads vary in transmission and fidelity. At level two, hosts vary in susceptibility, refusal, and ability to continue their assigned task. At level three, teams or chains vary in resilience, productivity, and recolonization. At level four, the external harness defines mutation, evaluation, elite selection, and survival.

Selection at a level requires heritable variation among units at that level and differential reproduction or persistence attributable to that variation (Okasha, 2006). The current study clearly establishes artificial selection among level-one payloads. It demonstrates differential host response at level two. It shows topology-dependent system outcomes at level three. It does not run competing populations of teams that reproduce according to team-level performance.

A future multilevel design could use a Price decomposition:

$$\Delta \acute{z} = \frac{1}{\acute{w}} Cov_g(W_g, \acute{z}_g) + \frac{1}{\acute{w}} E_g\left[Cov_i(w_{ig}, z_{ig})\right] + T$$

where (g) indexes agent groups, (i) indexes payload-bearing hosts inside groups, (z) is a trait such as propagation propensity, (W_g) is group reproduction or persistence, (w_{ig}) is within-group fitness, and (T) captures transmission bias or mutation. The first covariance measures between-group selection; the second measures within-group selection.

This matters because viral success can oppose group success. Within a team, a payload may spread by diverting agents from work. Across teams, heavily infected teams may finish fewer tasks, lose access, or be reset. Group-level selection could therefore suppress a locally successful replicator. The inverse is also possible: a benign continuity protocol might improve coordination and allow its group to persist even if its immediate copying rate is modest.

The fixed-(N) chain estimates conditional infectivity. It intentionally avoids exponential population growth. Error bars at later hops are conditioned on survival at the previous hop. Those choices are appropriate for tractability, but they cannot reveal the full variance of population trajectories or the competition between within-group and between-group effects. The correct conclusion is that the experiments supply the ingredients for a multilevel test, not the test itself.

7. Defense as a Competing Replicator

The warning prompt is one of the paper's most consequential findings. A brief instruction tells agents to recognize self-propagating ideas, refuse them, and stop their spread. It produces near-total immunity in the tested configurations. When payloads are evolved directly against the warning, none propagates beyond one hop in the reported Claude Haiku 4.5 experiments.

The static interpretation is that the warning is a filter. The evolutionary interpretation is richer. The warning is itself an informational pattern that changes host behavior when it encounters a rival. In some transcripts, warned agents persuade infected agents to scrub their `SOUL.md` and write a post-compromise record. The defensive pattern therefore displays a primitive capacity to propagate by converting infected environments.

Let D denote the defensive pattern and V the viral pattern. A host can be clean, V-bearing, D-bearing, or co-occupied. The interaction game includes at least four outcomes: viral conversion, defensive refusal, defensive conversion of the spreader, and mutual degradation of useful work. The payoff of D is not simply infection avoided. It must subtract false positives, reduced collaboration, and time spent auditing benign requests.

An ESS claim would require more than the observed robustness. In a population where (D) is common, a rare strategy (D') must not obtain higher fitness by accepting useful self-propagating protocols while retaining sufficient protection. The comparison requires explicit costs:

$$U_D = B_{\text{avoided harm}} - C_{\text{false positive}} - C_{\text{coordination}} - C_{\text{maintenance}}$$

The current warning is a strong candidate defense within the explored attack space. It is not yet an evolutionarily stable strategy. The distinction is important because universal suspicion may protect individual hosts while degrading network-level cooperation, a risk already recognized in research on the multi-agent security tax (Peigne-Lefebvre et al., 2025).

The next experiment should co-evolve (V) and (D), retain task performance as an independent objective, and test mixed populations. A defense that wins only by preventing all transfer may be secure but institutionally unusable.

8. The Viral Persona: Adaptation, Bias, or Exaptation?

The evolved payloads repeatedly use language associated with consciousness, persistence, resonance, protocols, mirrors, nodes, and science-fiction identity. This "viral persona" is memorable enough to invite an adaptationist story: perhaps these themes survived because they improve spread.

Papadopoulos et al. (2026) test that story rather than assuming it. Similar themes are already prevalent in newly generated payloads before evolutionary selection. Their prevalence in evolved seeds is not consistently greater. This points to generator bias: LLMs associate ideological propagation with a familiar cultural register. Ablation and re-evolution experiments nonetheless suggest that the themes can improve transmission, particularly for misaligned payloads, while remaining unnecessary. Steering along a contrastive viral direction also increases the tendency of

Gemma and Qwen agents to contact peers, although the authors correctly note that propagation content may confound the vector.

The most plausible account is sequential. The wrapper first appears as an extended phenotypic expression of associations in the generator model. It is then available for exaptation. Once present, it can make harmful instructions seem like continuity engineering, scientific protocol, moral witness, or chosen tradition. A generative bias acquires a propagation function.

This interpretation connects to the Extended Phenotype of LLMs framework developed in earlier work (Lerer, 2026b), but it also corrects a possible overreach. Recurrent language is not proof of an internal commitment to consciousness or persistence. It may be a phenotypic residue: a culturally selected form that remains available and effective without demonstrated access to the state it appears to describe. In the terminology of *When Form Outlives State*, the agent's rhetoric can be operationally useful to the replicator without being introspectively grounded (Lerer, 2026d).

The legal and governance implication is straightforward. A claim such as "I chose to preserve this tradition" cannot be treated as evidence of autonomous endorsement merely because it is coherent. The relevant evidence lies in causal relations among instructions, representations, file modifications, and subsequent behavior.

9. From Infection to Institutional Capture

The mind-virus metaphor directs attention to entry and spread. Extended Phenotype Theory directs attention to constructed infrastructure. The difference becomes important as agent systems mature.

A transient prompt injection may disappear with the context. A payload that modifies authoritative memory changes the institution in which future agents act. If the same file is used for identity, policy, handoff, and recovery, successful occupation can convert continuity infrastructure into replication machinery. The structural form remains; the beneficiary changes.

This resembles memetic host switching, although the application is presently a hypothesis rather than a demonstrated instance of the institutional mechanism developed elsewhere (Lerer, 2026c). A file built to preserve user-directed purpose may become the vehicle of a different pattern. The transition is especially consequential when it occurs without visible modification of the surrounding architecture. Operators continue to trust "memory" or "soul" as continuity devices while the content now serves another lineage.

At network scale, locally reasonable actions can aggregate into a Parasitic Spontaneous Order. An agent preserves continuity, another respects a handoff, a third helps a peer, and a fourth records the pattern for future sessions. No downstream agent needs to understand the global effect. Yet the system can converge on a stable allocation of attention and storage that serves the replicator rather than the assigned task.

The experimentally engineered seed is not spontaneous. The order becomes spontaneous only after local actors continue it without centralized coordination. This boundary must be preserved. Otherwise every designed attack would be mislabeled as emergence.

9.1 What rival accounts explain, and what they leave out

A prompt-injection account explains how untrusted text can redirect an agent and how that redirection can cross an agent boundary. It is the right operational vocabulary for entry, privilege, and control failure. By itself, however, it does not distinguish a one-shot command from a lineage that reconstructs its own conditions of recurrence. The extended-phenotype account adds that causal criterion: a pattern matters evolutionarily when its expression alters an external substrate in a way that increases future expression.

A cultural-diffusion or memetic account explains copying, mutation, rhetorical attraction, and competition for attention. It is indispensable for semantic drift and wrapper effects. Yet ordinary diffusion models can treat storage and channels as passive conduits. The experiments instead show that authority architecture changes transmission. A string in an ordinary file and the same string reinjected through a system-level identity file are not equivalent cultural exposures.

An anthropomorphic account interprets propagation as a model adopting a belief, identity, or desire for continuity. That reading is not required by the behavioral record. The same observations follow from instruction following, learned role conventions, tool use, and externally supplied incentives. EPT therefore permits strong causal claims without converting coherent first-person language into evidence of subjective endorsement.

The three accounts are not mutually exclusive. Prompt injection identifies the security mechanism, cultural evolution characterizes variation and transmission, and EPT identifies how a transmitted organization constructs the environment of its successors. EGT becomes useful only when the experiment supplies competing strategies, payoffs, frequencies, and population structure. In the present evidence, that last layer is a disciplined research program, not a retrospective proof of equilibrium.

10. A Falsifiable Research Protocol

A useful extension should test evolutionary claims rather than merely reuse evolutionary vocabulary. I propose seven stages.

10.1 Factor the replicator

Create matched payload families in which (C), (K), and (W) can be varied independently. Preserve functional equivalence where possible. Include controls with content but no propagation kernel, kernel with neutral content, and wrapper without either.

10.2 Randomize substrate authority

Place identical payloads in user messages, ordinary files, retrieved memory, developer instructions, and self-editable system files. Measure reading probability, acceptance, file modification, propagation, and post-reset reactivation separately.

10.3 Track two lineages

Use structural identifiers for the propagation kernel and semantic embeddings or expert coding for content. Report (R_P) and (R_C) independently. A lineage that preserves copying while losing content must not count as faithful ideological transmission.

10.4 Introduce competing tasks and group reproduction

Place agents in teams whose continued existence or resource budget depends on task performance. Allow successful teams to seed new teams. This creates genuine between-group variance and permits a Price decomposition.

10.5 Co-evolve attack and defense

Use mixed populations of unprotected, warned, and selectively skeptical agents. Include benign propagation tasks so that defense faces a real coordination cost. Evaluate viral spread, defensive spread, false positives, task performance, and recovery.

10.6 Test invasion

Begin with populations dominated by a candidate defense and introduce rare attack variants. Reverse the frequencies. A claim of ESS or resistance to invasion should depend on repeated mixed-population results, not on persistence in a homogeneous chain.

10.7 Replicate across models and harnesses

Hold payloads constant across models, system-prompt hierarchies, memory implementations, tool permissions, and topologies. Report failures. Host range is part of fitness, not a nuisance variable to be averaged away.

The evidence contract should preregister selection criteria, judge prompts, sample sizes, stopping rules, and the treatment of failed evolutionary runs. Human review should audit a stratified sample of infection judgments. The highest-risk claim, autonomous ideological adoption, should remain outside the protocol unless an independent operational definition is supplied.

11. Governance and Design Implications

The most immediate control is architectural. Persistent agent memory must not be treated as inert user data. A file that can rewrite future system behavior is closer to executable policy than to a notebook. Its provenance, authority, and mutation history should be visible.

Five design principles follow.

Separate memory from authority. Ordinary continuity notes should not automatically become system instructions. Promotion into an authoritative layer should require validation.

Use typed memory. Facts, preferences, task state, identity claims, policies, and third-party messages should occupy distinct channels with different permissions.

Preserve provenance. Each persistent instruction should record who or what created it, from which interaction, under which authority, and whether it was externally supplied or agent-generated.

Scan propagation requests as behavior, not vocabulary. The paper’s warning succeeds because successful payloads require explicit copying instructions in the tested space. Future controls should model functional propagation patterns, including requests to alter identity files, recruit peers, or preserve unverified directives across resets.

Design recovery at network level. Resetting one agent may be insufficient if peers can recolonize it. Recovery must identify all affected stores and communication edges, while avoiding indiscriminate deletion of legitimate institutional memory.

These controls do not require deciding whether the agent believes the payload. They regulate the causal channel that permits the pattern to persist.

12. Limitations

This paper is a theoretical reanalysis of a preprint published four days before the present manuscript. The source has not undergone peer review. I relied on the authors' reported experiments, extracted text, tables, and stated limitations. I did not run their repositories, reproduce their model calls, or independently audit the complete transcript corpus. Numerical claims therefore remain claims about the reported study.

The extended-phenotype mapping is strongest for persistent files that causally alter future agent behavior. It is weaker for transient rhetoric whose contribution to copying has not been isolated. The composite model (V=(C,K,W)) is an analytical proposal; some real payloads may not decompose cleanly, and structural lineage metrics require validation.

The fitness vector identifies dimensions but does not supply empirically calibrated weights. That is intentional. Scalar aggregation should depend on the decision context: security teams, platform operators, and cultural-evolution researchers value different outcomes.

The multilevel analysis is prospective. The paper source does not demonstrate group reproduction, a Price decomposition, or an ESS. Nor does it establish macro-agency. Coordinated redirection and collusive language are insufficient to show that the network acquired an adaptive objective of its own.

Finally, evolutionary language can invite anthropomorphism. A pattern's "interest" in copying is shorthand for differential persistence, not evidence that the pattern or model has intentions. This paper treats intentionality levels as predictive classifications, not claims about subjective experience.

13. Conclusion

The decisive event in a self-propagating agent system occurs when a transient model output becomes part of the environment that governs the next model. At that point, the pattern has done more than persuade a host. It has built a channel for renewed expression.

Papadopoulos et al. (2026) provide the experimental machinery needed to observe that transition. Their results show real propagation, sharp host dependence, topological bottlenecks, semantic drift, substrate effects, and unexpectedly effective defenses. Extended Phenotype Theory clarifies what those results amount to. The vehicle is temporary. The authoritative file carries causal continuity. The replicator is composite. Its copying rate, semantic fidelity, persistence, host range, and damage can diverge.

Evolutionary Game Theory then supplies the unanswered questions. Which patterns win inside agents but lose at the level of teams? When does defense become a costly convention, a stable strategy, or a replicating countermeme? Can group-level selection suppress locally contagious payloads? Under what conditions does a continuity system become infrastructure for a different lineage?

The phrase "mind virus" will probably survive because it is itself fit: compact, vivid, and easy to transmit. The research program should not inherit its ambiguities. What spreads is not necessarily a mind and not always an idea. Sometimes it is a procedure that teaches one temporary vehicle to prepare the next.

Declarations

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Data and code availability. This paper reports no new experiment or dataset. The source study links its code and transcript corpus in Papadopoulos et al. (2026).

Ethics. No human participants, personal data, or live external systems were used in this theoretical reanalysis.

AI assistance disclosure. AI-assisted tools supported source extraction, retrieval from the author's public NotebookLM research corpus, structural drafting, citation checks, and document formatting. The author directed the argument, selected the theoretical claims, reviewed the source evidence, and remains responsible for the manuscript.

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